

How and when AI-driven capabilities foster organizational sustainability: A moderated mediation analysis of digital literacy and green HRM

Adnan Ali^{a,*}, Qian Yang^a, Afzaal Ali^b, Zeeshan Ali^c, Muhammad Noman^d

^a School of Management, Northwestern Polytechnical University, 127 West Youyi Road, Beilin District, Xi'an Shaanxi, 710072, P.R.China

^b School of Management and Economics, University of Electronic Science and Technology of China, No.4, Section 2, North Jianshe Road, 610054, Sichuan, China

^c School of Economics and Management, China Three Gorges University, Yichang, Hubei 443002, China

^d School of Management, Zhengzhou University, Zhengzhou, Henan 45001, China

ARTICLE INFO

Keywords:

AI-driven capabilities
Dynamic capability view
Organizational sustainability
Green human resource management
Digital literacy

ABSTRACT

Researchers and practitioners widely view artificial intelligence-driven (AI) capabilities as the future of transforming organizations toward sustainability and making the world more sustainable. However, their crucial role and the specific mechanisms through which they influence organizational sustainability (OS) across three key dimensions—economic, environmental, and social performance—remain underexplored in the literature. To address this gap, this study adopting dynamic capability theory proposes a moderated mediation (MODMED) model that investigates the relationship between AI competencies OS sustainability and how this relationship is mediated by green human resource management (GHRM) and moderated by digital literacy. Using a sample of 310 Chinese firms, our results show the positive influence of AI capabilities on OS and the mediating role of GHRM. The MODMED analysis further reveals that digital literacy strengthens the indirect link between AI capabilities and OS via GHRM. This study offers fresh insights into how firms can effectively leverage AI capabilities, with the support of GHRM and digital literacy to drive OS.

1. Introduction

In response to ongoing environmental degradation and rising stakeholder demands for sustainable practices, organizations face increasing pressure to act ethically and incorporate sustainability into their operations and long-term strategies (Uddin et al., 2023). Organizational sustainability (OS) refers to a firm's ability to manage its environmental, social, and economic impacts while ensuring long-standing value creation for both the business and society (Bianchi et al., 2022). Firms that adopt sustainable practices focus on minimizing resource consumption, reducing waste, and aligning their operations with societal and environmental expectations (Ortiz-Avram et al., 2024), which enhances their resilience and adaptability to regulatory and market changes (Mazutis et al., 2021). Moreover, OS strengthens relationships with customers, investors, and employees, ultimately boosting competitive advantage and market reputation (Fan et al., 2021). In the current competitive landscape, sustainability drives innovation, operational efficiency, and future growth (Bocken & Geradts, 2020). Therefore, it becomes essential for firms to deeply embed

sustainability into their business models to ensure long-term competitiveness and contribute to a more sustainable world.

To implement OS effectively, research indicates that firms must cultivate robust artificial intelligence-driven (AI) capabilities (Abulibdeh et al., 2024; Ayoub & Aljuhmani, 2024; Kar et al., 2022). Drawing on dynamic capability view (DCV), AI-based technologies—like ChatGPT, the Internet of Things (IoT), blockchain, cloud computing, machine learning, and big data (Qamar et al., 2021; Roppelt et al., 2025)—enhance a firm's ability to adjust, integrate, and reorganize internal and external resources to tackle new sustainability challenges that include environmental, social, and economic dimensions (Pan & Nishant, 2023). These technologies allow firms to process extensive data, optimize resources, and monitor their operations' environmental and social impact, leading to more informed decision-making aligned with sustainability goals (Liao et al., 2022). Additionally, AI-driven insights enable firms more resilient and adaptive in the changing sustainability landscape by enabling them to predict and react to changes in market trends, customer preferences, and regulations (Benzidia et al., 2021; Dohale et al., 2024). Ultimately, AI technologies empower firms to

* Corresponding author.

E-mail addresses: adnan268010@yahoo.com (A. Ali), qianyang@nwpu.edu.cn (Q. Yang), afzaalali.iui@gmail.com (A. Ali), zeeshan.ali1992@ctgu.edu.cn (Z. Ali), nomanmuhammad@stu.xjtu.edu.cn (M. Noman).

<https://doi.org/10.1016/j.ijinfomgt.2025.102956>

Received 11 April 2025; Received in revised form 27 April 2025; Accepted 15 July 2025

Available online 19 July 2025

0268-4012/© 2025 Elsevier Ltd. All rights reserved, including those for text and data mining, AI training, and similar technologies.

not only meet sustainability targets but also to innovate and enhance their long-term competitiveness (Ogbeibu et al., 2024). Despite the growing recognition of AI's role in enhancing organizational innovation and operational efficiency (Shaik et al., 2023), its specific contribution to promoting OS remains underexplored in current literature. To address this gap, this study investigates the first key question: *How do AI capabilities promote OS?*

Moreover, the existing literature offers limited clarity on how AI capabilities influence OS, leaving the mediating processes underexplored. Green human resource management (GHRM), which integrates sustainability concepts into HR practices and policies (Úbeda-García et al., 2021)—emerges as a key strategic tool that helps fill this gap and allows firms to fully utilize AI's potential for sustainability. Effective implementation of OS also requires a green culture that embeds sustainability into organizational values, practices, and employee engagement strategies (Jerónimo et al., 2020). By encouraging responsible behavior, developing pertinent skills, and fostering sustainable attitudes through hiring, training, performance management, and rewards, GHRM enables employees suitable with the firm's sustainability goals (Jerónimo et al., 2020). This alignment maximizes the impact of AI technologies on sustainability outcomes. By cultivating a sustainability-oriented culture, GHRM practices enable firms to enhance the effectiveness of AI-driven technologies, reducing environmental and social impacts while improving economic efficiency (Hao et al., 2024). In line with the dynamic capability view (Teece et al., 2016), AI capabilities further enhance GHRM by providing data-driven insights, automating sustainable workflows, and fostering collaboration, enabling organizations to achieve their sustainability objectives more effectively (Dey et al., 2024). Regardless of this, there hasn't been much focus in the literature on GHRM's mediating function in connecting AI capabilities with OS. This work aims to fill this gap by answering the second research question, *"How does GHRM mediate the link between AI capabilities and OS?"*

Furthermore, while AI capabilities are essential for advancing GHRM and OS, the extent to which firms can achieve superior sustainability outcomes may hinge on contextual factors that either amplify or diminish these indirect effects (Liu et al., 2023; Nishant et al., 2020). This study highlights digital literacy—defined as employees' proficiency with digital tools and platforms (Nikou et al., 2022)—as a crucial moderator, reflecting how effectively employees can interact with and apply AI-enabled technologies (Cetindamar et al., 2021b). Since leveraging AI for GHRM requires digital adeptness (Ogbeibu et al., 2024), digital literacy enables employees to integrate AI-driven sustainability practices more seamlessly into their workflows. High digital literacy empowers employees to interpret and act on AI-generated insights more precisely, promoting consistency in sustainable practices and resource management (Zechiel et al., 2024). Additionally, digitally literate employees engage in continuous learning and can respond more agilely to AI-driven sustainability goals (Abulibdeh et al., 2024), thereby

reinforcing the alignment between GHRM and OS. By considering both mediating and moderating effects (Preacher et al., 2007), the study offers an enriched inquiry into the connection between AI capabilities and OS, emphasizing the mediating role of GHRM and the moderating influence of digital literacy. Despite the importance of digital literacy in facilitating AI-driven initiatives, current literature lacks clarity on its moderating role in the indirect link between AI capabilities and OS through GHRM. Consequently, this study addresses this gap by examining the third research question: *How does digital literacy moderate the indirect relationship between AI capabilities and OS via GHRM?*

In a nutshell, our study contributes to the current stream of digital and sustainability literature in three distinct ways by presenting a moderated mediation (MODMED) model (see Fig. 1). First, it expands the theoretical and empirical knowledge on the connection between OS and AI capabilities. While existing research acknowledges AI's role in enhancing firm performance (Shaik et al., 2023), its specific contribution to promoting sustainability across economic, environmental, and social dimensions remains underexplored. Second, it enriches the scholarship by bridging the missing link between AI capabilities and OS by highlighting the mediating role of GHRM. By integrating GHRM as a key mechanism, it offers a comprehensive perspective on AI's role in driving sustainable outcomes, thereby providing a more holistic view of AI's impact on OS. Finally, by incorporating digital literacy as a contingent factor, the study advances our understanding of how and under what conditions AI capabilities exert a stronger influence on OS via GHRM. This MODMED model (Preacher et al., 2007) highlights a comprehensive view of the boundary conditions that shape the pathway from AI capabilities to sustainable outcomes.

2. Theory and hypotheses developments

2.1. Organizational sustainability

OS emphasizes long-term resilience by integrating economic, environmental, and social goals into a firm's operations and decision-making processes procedures (Negri et al., 2021). Organizations committed to sustainability go beyond profit-seeking, actively considering the influence of their actions on the environment, society, and future generations (Fischer et al., 2020). They adopt practices that minimize waste, reduce carbon emissions, and use resources responsibly, often promoting a circular economy approach. In addition, they prioritize social responsibility by respecting employees, customers, and communities and investing in human capital to foster innovation and social well-being (Ishaq et al., 2024). By balancing these dimensions, organizations committed to sustainability enhance their adaptability and build trust, contributing positively to society and strengthening their position in a sustainability-oriented market.

Existing studies have identified various antecedents to OS. For instance, Smith (2012) highlights organizational learning as a crucial

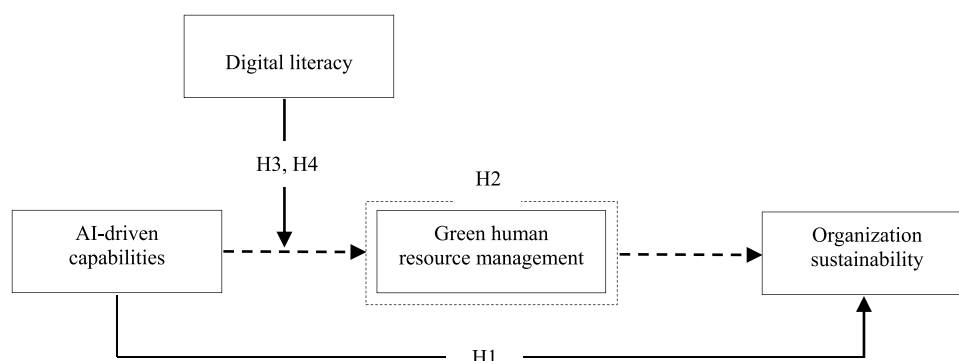


Fig. 1. Conceptual model. Note: Dotted arrows show mediation relationship.

factor for sustainability, while Carayannis et al. (2015) underscore the role of revolutionizing business models. Similarly, Jerónimo et al. (2020) point to GHRM practices as essential contributors to sustainability, and Demir et al. (2023) find knowledge management to positively impacts sustainable outcomes. Xie and Zhu (2020) further reveal that green training initiatives can promote sustainability. Even though these studies offer insightful information, more research is needed to fully understand the impact of AI capabilities as a critical antecedent to OS.

2.2. AI capabilities and sustainable organization: dynamic capability view

The DCV suggests that an organization's success in dynamic environments relies on its ability to innovate, integrate, and reorganize resources and capabilities (Gupta et al., 2020; Teece et al., 1997). Dissimilar to static capabilities, which emphasize routine tasks, dynamic competences stress significance of detecting opportunities, grabbing them, and renovating processes to address new challenges (Stoyanova & Stoyanov, 2024). This approach is especially critical in volatile markets, where organizational flexibility and innovation determine sustained competitive advantage. For the effective implementation of OS, dynamic capabilities have emerged as strategic assets that permit companies to continuously familiarize to evolving environmental, social, and economic requirements. Chari et al. (2022) emphasize that these dynamic capabilities empower firms to swiftly respond to shifting market conditions while actively driving sustainability initiatives. Dynamic capabilities, organizations can effectively integrate sustainable practices across all functions, enabling them to address stakeholder pressures and respond to long-term sustainability challenges with increased agility and resilience (Yi & Demirel, 2023).

When viewed through the lens of DCV, AI capabilities emerge as a critical dynamic capability that significantly influences OS (Bedué & Fritzsche, 2022). By exploiting AI, firms can collect and scrutinize extensive information sets to uncover emerging sustainability opportunities, optimize resource utilization, and identify potential risks that could jeopardize their sustainability objectives (Pan & Nishant, 2023). For instance, AI-driven insights enable firms to predict environmental impacts, enhance energy efficiency, and minimize waste, thereby improving their environmental performance (Singh et al., 2024). Economically, AI enhances operating efficiency, decreases expenditures, and assists decisions based on data analysis (Hao & Demir, 2024), all adding to sustainable development (Czarnitzki et al., 2023). On the social front, AI capabilities promote employee well-being by encouraging ethical work practices, supporting equitable hiring processes, fostering diversity, and offering personalized employee development programs (Chang & Ke, 2024). Consequently, AI has appeared as a strategic instrument empowering firms to adopt OS as a means to remain competitive while fulfilling their corporate responsibilities in the current complex and rapidly evolving business environment.

2.3. AI capabilities and organizational sustainability

In line with the DCV, AI-driven capabilities are crucial assets that strengthen an organization's ability to achieve and sustain multidimensional sustainability. The DCV emphasizes adaptability

through the ongoing renewal and integration of resources to meet evolving demands (Ortiz-Avram et al., 2024). AI is renovating the environmental dimension of sustainability by enhancing resource management, reducing emissions, and minimizing waste (Vaio et al., 2020). Benzidia et al. (2021) argue that advanced data analysis enables the real-time processing of massive volumes of environmental information, offering intuitions into the ecological impacts of various operations. For instance, AI-powered predictive models empower firms to anticipate emissions, optimize energy consumption, and minimize resource waste (Ahmad et al., 2021). By leveraging AI to assess their environmental footprint, firms can proactively adjust production

processes, logistics, and supply chain practices. These adjustments promote eco-friendly initiatives beyond regulatory compliance to enhance environmental performance.

Furthermore, AI capabilities significantly enhance the economic dimension of OS by optimizing operations, reducing costs, and aligning decision-making with sustainable growth. For instance, AI-driven demand forecasting enables firms to manage inventory with precision, minimizing excess stock and potential shortages while also lowering holding costs (Jauhar et al., 2024). According to Reyes et al. (2023), predictive analytics streamline production and supply chain management, helping companies avoid disruptions and improving their responsiveness to shifting market demands. Additionally, Latinovic and Chatterjee (2022) highlight that AI insights empower firms to identify emerging customer needs, analyze market trends, and uncover cost-saving opportunities. These proactive adjustments ensure not only financial resilience but also alignment with sustainability goals. By leveraging these efficiencies, AI supports organizations in achieving cost-effective and sustainable growth within a competitive landscape.

On the social front, AI capabilities are essential in creating a workplace that promote

fairness and ethical behaviors. Malik et al. (2023) demonstrate that AI can help design equitable hiring processes, bolster diversity initiatives, and promote employee well-being through customized training and development programs. Furthermore, Budhwar et al. (2022) emphasize that AI-driven systems enable organizations to analyze employee feedback and performance data, facilitating the creation of a supportive work culture that addresses individual needs and ensures equitable opportunities. This strategic application of AI augments employee experience and loyalty and adds positively to OS by nurturing a stable and motivated workforce (Kumar et al., 2024). Consequently, AI capabilities enhance the social aspect of sustainability by promoting ethical behavior and fostering an inclusive workplace that supports both employees and the broader community.

H1. : AI-driven capabilities are positively linked with organizational sustainability.

2.4. The mediating role of green HRM

GHRM practices incorporate socially responsible principles into HR functions to support the firms' broader sustainability efforts (Ogbeibu et al., 2024). These practices include "green recruitment," "training," "performance management," "rewards," and "employee engagement" aimed at aligning the workforce with firms' sustainability purposes (Yong et al., 2020). For instance, green recruitment involves selecting candidates who value sustainability, while green training and development focus on building employees' eco-friendly skills and behaviors (Usman et al., 2023). Performance management in GHRM recognizes and rewards sustainable practices among employees, reinforcing environmentally friendly behavior throughout the organization (Al-Swidi et al., 2021). Together, these practices foster a workplace that promotes sustainable actions and integrates green values into the organizational culture. This approach not only drives individual green behaviors among employees but also fosters a collective commitment to sustainable impact.

We broaden the existing stream of scholarship by postulating that GHRM assists as a central function in translating the influence of AI-driven capabilities into tangible OS by aligning employee actions, practices, and mindsets with the firm's sustainability goals. While AI-driven capabilities provide organizations with advanced tools to monitor, analyze, and optimize operations across "environmental," "social," and "economic" dimensions (Lee et al., 2024), their successful implementation often depends on workforce engagement and adherence to sustainable practices (Di Vaio et al., 2020). GHRM expedites this alignment by embedding sustainability objectives within HR functions—e.g., green staffing, training, and performance

supervision—ensuring that employees possess the essential competencies and are motivated to actively champion sustainable initiatives (Ogbeibu et al., 2024). For instance, AI-powered insights can guide GHRM in designing training modules that focus on sustainable resource management and eco-conscious decision-making, allowing employees to integrate environmentally responsible behaviors more effectively in their daily tasks (Hao et al., 2024). By incorporating sustainability-focused goals into HR processes, GHRM helps to bridge AI insights with practical applications, enabling AI-driven sustainability initiatives to become more accessible and actionable for employees (Chang & Ke, 2024).

Furthermore, GHRM amplifies the impact of AI-driven capabilities by fostering a sustainability-oriented organizational culture that actively supports the firm's broader ecological, social, and financial objectives. By incorporating sustainability achievements into performance evaluations and aligning incentive systems with sustainability milestones, GHRM encourages employees to apply AI insights in ways that drive OS (Trujillo-Gallego et al., 2022). For example, rewards and recognition for contributions to sustainability help establish an environment where AI-powered initiatives—like optimizing resource use, reducing waste, and promoting social responsibility—are fully embraced by the workforce (de Oliveira et al., 2023). This alignment fosters a shared commitment throughout the organization, motivating employees to take personal responsibility and actively contribute to the organization's sustainability objectives. In this regard, GHRM functions as a critical mediator, transforming the potential of AI-driven capabilities into sustainable practices that augment the firm's ecological, social, and economic outcomes.

H2. : Green human resource management practices mediate the link between AI-driven capabilities and organizational sustainability.

2.5. The moderating role of digital literacy

Digital literacy encompasses the competencies and understanding required to effectively utilize digital technologies, tools, and platforms (Nikou et al., 2022). It encompasses various competencies, such as the capacity to search for, assess, and apply information online, as well as proficiency in using digital tools for communication, collaboration, and problem-solving (Cetindamar et al., 2021b). It also includes understanding security, privacy, and data safety, empowering individuals to navigate potential risks in the digital setting (Irfan et al., 2022). In the workplace, digital literacy enables employees to interact efficiently with new technologies, adapt to changing digital landscapes, and leverage digital tools for productivity and innovation. High levels of digital literacy empower individuals not only to use digital tools proficiently but also to comprehend the ethical implications and security risks related to digital data, promoting a responsible and informed use of technology.

We suggest that digital literacy may moderate the link between AI capabilities and GHRM by enhancing employees' ability to adopt and support AI-driven sustainability initiatives. When employees have strong digital literacy skills, they are more capable of understanding, engaging with, and incorporating AI tools into GHRM practices (Ogbeibu et al., 2024). For instance, digitally literate employees can use AI-powered platforms to track and report sustainable practices, engage with environmental metrics, and support GHRM initiatives like green training programs and efficient resource management (Chin et al., 2024). This enhanced capability allows them to fully leverage AI insights, driving more meaningful participation in the organization's green initiatives. Conversely, when digital literacy is low, employees may struggle to engage with AI-driven processes, reducing the impact of AI technologies in promoting sustainability through GHRM. Therefore, digital literacy strengthens a firm's AI capabilities in fostering GHRM by equipping employees to use technology more effectively, creating a robust foundation for advancing OS goals.

H3. : Digital literacy moderates the link between AI-driven capabilities

and green human resource management practices.

We further expect that digital literacy moderates the indirect relationship between AI-driven capabilities and OS through GHRM, with the effect being stronger when digital literacy is high compared to when it is low (Preacher et al., 2007). Employees with strong digital literacy skills may engage more effectively with AI-powered GHRM tools, such as platforms for monitoring resource usage, tracking carbon footprints, or measuring the impact of sustainable activities (Ogbeibu et al., 2024). This familiarity allows employees to better interpret and apply AI-driven insights, making GHRM practices more impactful and ensuring that sustainability initiatives are well-integrated throughout the organization (Alnasser et al., 2024). For example, digital literacy enables employees to participate fully in eco-friendly training programs that use AI analytics to personalize learning (Kılıç & Atilla, 2024), promoting socially responsible behavior that supports the organization's sustainability objectives (Abulibdeh et al., 2024).

In contrast, when digital literacy levels are low, employees may struggle to utilize AI tools within GHRM practices, weakening AI capabilities' effect on sustainability outcomes. This gap can reduce the effectiveness of GHRM initiatives, as employees may find it challenging to interpret AI insights or implement AI-driven recommendations without adequate digital skills (Chowdhury et al., 2023). As a result, digital literacy acts as a critical factor, amplifying the extent to which AI capabilities—channeled through GHRM practices—contribute to OS. Ultimately, when employees are digitally literate, they can effectively leverage AI tools within GHRM (Trujillo-Gallego et al., 2022), which improves the organization's capacity to meet sustainability objectives across environmental, economic, and social dimensions.

In summary, the arguments presented support a MODMED effect of digital literacy, emphasizing its crucial role in enhancing the indirect impact of AI technologies on OS through GHRM. This study offers a deeper understanding of the relationship between AI capabilities and OS, with a focus on the mediating role of GHRM and the moderating effect of digital literacy.

H4. : The indirect relationship between AI-driven capabilities and organizational sustainability via green human resource management practices is stronger under high digital literacy than low.

3. Methodology

3.1. Sample and data collection

To collect data for this study, we surveyed manufacturing firms throughout China. The questionnaire was initially developed in English, drawing on established research and insights from in-depth interviews. For linguistic and cultural accuracy, two experts in strategic management translated the questionnaire into Chinese, and two additional scholars performed a back-translation into English to ensure it aligned closely with the original content (Berry, 1980). To validate its content, we conducted a pilot study involving twenty CEOs from ten local firms, whose feedback allowed us to refine the Chinese version before administering the final survey through face-to-face interviews.

To ensure a balanced regional representation across China's eastern, central, and western areas (distributed over 23 of 31 provinces), we selected 500 firms from each region, resulting in a comprehensive sample of 1500 firms based on information from local commerce departments. Professional interviewers contacted senior managers via phone or email, outlining the survey's objectives and verifying essential company information. Of the initial contact, 340 firms agreed to participate in our study. Follow-up calls were made to coordinate interview schedules and finalize data collection logistics.

Five trained interviewers conducted face-to-face interviews, gathering in-depth data from senior managers involved in sustainability and digital strategy decision-making. This approach helped improve response accuracy and expedite data collection. To minimize common

method bias (CMB), we collected information from two senior respondents per firm, such as chairpersons, CEOs, and key managers, while assuring them of confidentiality and that there were no “right” or “wrong” answers to reduce social desirability bias. Chairpersons and CEOs provided insights into AI capabilities and sustainability practices, while senior managers offered details on digital literacy, GHRM, and other control variables. Interviewers remained present during each session to clarify any questions, with interviews averaging 45 min. Excluding incomplete responses, we obtained 620 valid responses, representing 310 firms, with regional participation as follows: 42.25 % from the eastern region, 30.00 % from the central region, and 27.75 % from the western region. Appendix A presents the profile of the sample firms.

3.2. Variables measurement

All measures used in this study were adapted from recognized research, with each main construct (except for most control variables) assessed using a seven-point Likert scale. Detailed measurements for the primary constructs are provided in the Appendix.

3.2.1. Independent variable

We measured AI-driven capabilities using a five-item scale adapted from Belhadi et al. (2024) and Wamba (2022). This scale assesses a firm’s readiness with infrastructure and skilled resources for AI, its use of AI for forecasting and self-learning models, integration of AI across business units, and AI’s role in supporting new product and service development.

3.2.2. Mediating variable

In alignment with prior studies (Ishaq et al., 2024; Yong et al., 2020), GHRM is measured through six dimensions: “green analysis and job description (three items),” “green recruitment (two items),” “green selection (two items),” “green training (three items),” “green performance assessment (three items),” and “green rewards (two items).” These dimensions collectively capture various aspects of responsible HR practices, emphasizing the importance of integrating sustainability into every stage of the employee lifecycle.

3.2.3. Moderating variable

Following previous studies (Vuorikari et al., 2016; Zahoor et al., 2023), digital literacy is measured through five dimensions, namely “information literacy,” “interaction and collaboration,” “digital content creation,” “safety,” and “problem-solving.” Information literacy consists of three items that assess “confidence in browsing,” “evaluating,” and “managing digital data and content.” Interaction and collaboration include four items measuring the ability to “interact,” “share,” “engage in digital citizenship,” and “collaborate through digital technologies.” Digital content creation comprises three items that capture competence in “developing,” “integrating,” and “respecting copyright for digital content.” Safety includes four items focused on “awareness of protecting devices,” “personal privacy,” “health,” and “the environment in digital contexts”. Finally, problem-solving consists of four items that evaluate the “ability to address technical issues,” “identify needs,” “use technology creatively,” and “recognize digital skill gaps.”

3.2.4. Dependent variable

In line with previous studies (Laosirihongthong et al., 2013; Yong et al., 2020), OS is measured across three key dimensions—economic, environmental, and social performance—each represented by five items. Economic performance measures cost reductions and financial improvements, such as savings on materials, energy, and waste fees. Environmental performance assesses efforts to reduce emissions, waste, and hazardous materials while enhancing energy conservation. Social performance evaluates contributions to community welfare, employee safety, and protection of public rights.

3.2.5. Control variables

Following previous research, we control for competitive intensity, firm age, firm size, and state ownership to account for factors that may influence study outcomes. Specifically, firm age is calculated as the logarithm of years since establishment, and firm size is measured by the logarithm of workforce size. Additionally, state ownership is represented by a dummy variable, coding state-owned enterprises as 1 and non-state-owned enterprises as 0, to account for its potential impact on firm operations. Competitive intensity is measured on a 4-point scale, with values ranging from 1 (low) to 4 (exceptionally high).

3.3. Construct validity

We used SPSS 24 to evaluate the reliability and validity of the measures, following Anderson and Gerbing (1988). First, results for internal reliability, as detailed in Appendix B, show that Cronbach’s alpha values ranged from 0.76 to 0.89, all exceeding the recommended threshold of 0.7 (Hair et al., 2014). Second, to assess the validity of the latent scales, we conducted a four-factor confirmatory factor analysis (CFA) using covariance-based structural equation modeling with AMOS 24. Each measurement item was linked to its respective construct, with the covariances between constructs freely estimated. The CFA results showed that the measurement model fit the data well, as indicated by the following fit indices: $\chi^2/df = 2.315$ ($p < 0.001$), GFI = 0.947, CFI = 0.961, NNFI = 0.957, TLI = 0.916, SRMR = 0.053, and RMSEA = 0.068. These values suggest that the model adequately fits the data (Hair et al., 1998).

Furthermore, as detailed in Appendix B, all factor loadings obtained through principal components analysis with varimax rotation in SPSS 24, were significant and exceeded 0.7 (Comrey & Lee, 1992). Additionally, all average variance extracted (AVE) values surpassed the 0.5 threshold, and composite reliabilities exceeded 0.7 (Hair et al., 2014). To confirm discriminant validity, we performed a series of chi-square difference tests on each pair of latent variables. By comparing the chi-square values from constrained models, where correlations between paired constructs were fixed at 1.0, with those from unconstrained models, where correlations were freely estimated, we found that all tests were significant (e.g., AI capabilities vs. GHRM; $\chi^2(1) = 7.842$, $p < 0.001$), indicating sufficient discriminant validity (Anderson & Gerbing, 1988). Additionally, discriminant validity was further supported as the shared variance between each construct pair was less than the AVE for the individual constructs (Fornell & Larcker, 1981). Table I shows that the square roots of the AVE values (diagonal elements) for each construct exceed the off-diagonal elements, reinforcing discriminant validity.

3.4. Common method bias

To assess and mitigate potential CMB in our data, we employed several approaches. First, we collected data from different informants for the dependent and independent variables, as a multi-informant approach can reduce CMB risk (Jiang et al., 2021). Second, we conducted Harman’s single-factor test using CFA (Podsakoff, 2003). The analysis identified four factors with eigenvalues above 1.0, accounting for 68.48 % of the total variance, with the largest factor explaining 26.14 %—well below the critical 50 % threshold (Podsakoff, 2003). Third, we applied a marker variable (MV) test, following Lindell and Whitney (2001). We selected job satisfaction as the MV, which reflects employee contentment (Chatzopoulou et al., 2022) and is theoretically unrelated to the study’s constructs. Using the lowest positive correlation ($r = 0.002$) between the MV and other variables to adjust construct correlations, we confirmed that all latent variable correlations remained statistically significant. These results suggest that CMB is unlikely to pose a significant concern in our study.

Table 1
Descriptive statistics and correlations.

Main variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
1. Competitive intensity	1																		
2. Firm age ^a	0.002	1																	
3. Firm size ^a	0.004	0.014*	1																
4. State ownership	-0.002	-0.007	0.010	1															
5. AI-driven capabilities	0.192*	-0.005	-0.012	-0.017	0.871	1													
6. DL information literacy	0.014	0.030	-0.014	-0.155*	0.210**	0.854	1												
7. DL interaction and collaboration	0.012	0.021	0.004	0.010	0.251**	0.301***	0.863	1											
8. DL digital content creation	-0.011	-0.014	-0.003	0.001	0.254**	0.254**	0.844	0.891	1										
9. DL safety	-0.004	-0.006	-0.012	-0.101	0.305***	0.211**	0.241**	0.248**	0.891	1									
10. DL problem-solving	0.021	0.002	0.005	-0.015	0.281**	0.189*	0.174*	0.169*	0.210**	0.893	1								
11. GHRM green analysis and job description	-0.011	-0.020	0.012	-0.010	0.351***	0.259**	0.182*	0.201**	0.135*	0.187*	0.851	1							
12. GHRM green recruitment	0.005	0.006	0.001	0.020	0.311***	0.235**	0.231**	0.204**	0.267**	0.194*	0.189*	0.839	1						
13. GHRM selection	0.010	0.005	0.015	0.007	0.258**	0.261**	0.274**	0.221**	0.234**	0.278**	0.233*	0.274**	0.870	1					
14. GHRM training	-0.010	-0.110	-0.014	0.100	0.268**	0.268**	0.234**	0.277**	0.268**	0.274**	0.310**	0.318**	0.301***	0.902	1				
15. GHRM green performance assessment	0.021	0.011	0.005	0.008	0.355**	0.331**	0.247**	0.236**	0.270**	0.272**	0.172*	0.161*	0.210**	0.282**	0.866	1			
16. GHRM green rewards	-0.020	-0.015	-0.022	-0.011	0.287**	0.221**	0.195**	0.224**	0.226**	0.298**	0.211**	0.311**	0.341***	0.282**	0.252**	0.911	1		
17. OS economic performance	0.121*	0.132*	-0.114*	-0.221**	0.458***	0.310***	0.179*	0.189*	0.169*	0.247**	0.225**	0.223**	0.245**	0.269**	0.247**	0.264**	0.871	1	
18. OS environmental performance	0.110*	-0.102*	-0.023	0.051	0.335***	0.326***	0.161*	0.169*	0.124*	0.168*	0.185*	0.178*	0.199**	0.258**	0.247**	0.278**	0.471***	0.879	1
19. OS social performance	0.132*	-0.014	-0.011	-0.005	0.335***	0.347**	0.184*	0.247**	0.268**	0.220**	0.247**	0.168*	0.318***	0.299**	0.341**	0.324**	0.374***	0.441***	0.875
Mean	2.541	2.452	1.462	1.621	5.234	4.684	4.057	5.214	4.689	4.012	4.892	5.229	5.128	4.985	5.181	4.552	5.061	4.668	5.275
S.D.	0.485	0.654	0.584	0.754	0.948	0.785	0.641	0.628	0.568	0.874	0.759	0.837	0.713	0.847	0.680	0.481	0.696	0.920	0.594

Notes: *Log taken; ** $p < 0.05$; *** $p < 0.01$; 2-tailed; Diagonal elements (in bold) are square roots of the AVE values; N = 310
 DL = digital literacy, OS = organizational sustainability.

4. Results

Table I provides an overview of the variables' means, standard deviations, and correlations. Table II presents the results of the moderated hierarchical regression analysis. Models 1 and 4 include only control variables as baseline models. In Models 2 and 3, we incorporated AI capabilities and the interaction term between AI capabilities and digital literacy. Similarly, Models 5 and 6 introduce AI capabilities alongside the combined effect of AI capabilities and GHRM.

4.1. Main effect testing

H1 posits a positive relationship between AI capabilities and OS. The results presented in Model 5 of Table II indicates that AI capabilities significantly influence OS ($\beta = 0.254$, $p < 0.001$), thereby supporting H1.

4.2. Mediating effect testing

H2 proposes that GHRM mediates the link between AI capabilities and OS. Following Baron and Kenny (1986) approach to test this hypothesis, we assessed four conditions. First, as shown in Model 2 of Table II, AI capabilities significantly influence GHRM ($\beta = 0.184$, $p < 0.01$), fulfilling Condition 1. Second, in Model 5, AI capabilities show a significant positive effect on OS ($\beta = 0.254$, $p < 0.001$), meeting Condition 2. Third, Model 6 demonstrates that GHRM significantly affects OS ($\beta = 0.301$, $p < 0.001$), satisfying Condition 3. Finally, in Model 6, when GHRM is included, the effect of AI capabilities on OS decreases significantly ($\beta = 0.136$, $p < 0.05$), fulfilling Condition 4. These results align with Baron and Kenny's (1986) criteria, indicating that GHRM partially mediates the link between AI capabilities and OS, thereby supporting H2.

To further confirm these findings, we conducted both the Sobel test and the Monte Carlo Method for Assessing Mediation (MCMAM). The Sobel test results ($z = 2.34$, $p < 0.01$) affirm GHRM's mediating role and the MCMAM analysis, using Selig and Preacher's (2008) method

with a 95 % confidence interval of 0.068–0.276, further supports this mediation effect.

4.3. Moderated mediating effect testing

H3 posits that digital literacy moderates the relationship between AI capabilities and GHRM. To test this, we conducted hierarchical moderated regression analyses, applying mean-centering to reduce multicollinearity Aiken and West (1991). Table II presents the results, showing a statistically significant and positive interaction between AI capabilities and digital literacy ($\beta = 0.270$, $p < 0.001$), supporting H3. Using a simple slope test as suggested by Aiken and West (1991), we examined the interaction effect at high (+1 SD) and low (-1 SD) levels of digital literacy. Fig. 2 illustrates that the association between AI capabilities and GHRM strengthens more under high digital literacy compared to low digital literacy.

H4 proposes that digital literacy moderates the indirect effect of AI capabilities on OS via GHRM. To test this, we conducted a parametric bootstrap analysis using the MCMAM, as recommended by Selig and Preachers (2008), via the PROCESS macro. This analysis assesses the conditional indirect effect of AI capabilities on OS through GHRM across different levels of digital literacy. As shown in Table III, a 5,000-replication bootstrap sample with a 95 % confidence interval (CI) confirms a conditional indirect effect (Cheung & Lau, 2017), accepting H4. Specifically, the indirect effect of AI capabilities on OS through GHRM is substantially stronger under high digital literacy ($\beta = 0.215$, 95 % CI = [0.069, 0.364]) than under low digital literacy ($\beta = 0.039$, 95 % CI = [-0.033, 0.125]). These findings align with H4, highlighting digital literacy's role in moderating this indirect relationship.

Table II
Regression results.

Control variables	GHRM			Organizational sustainability		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Firm age ^e	0.021 (0.103)	0.026 (0.101)	0.025 (0.101)	0.104 (0.210)	0.111 (0.213)	0.115 (0.214)
Firm size ^e	-0.113* (0.011)	-0.115* (0.012)	-0.126* (0.023)	-0.024 (0.231)	- 0.025 (0.235)	- 0.027 (0.241)
Competitive intensity	0.125* (0.024)	0.123* (0.026)	0.121* (0.023)	0.172* (0.012)	0.153* (0.016)	0.162* (0.021)
State ownership	0.012 (0.204)	0.013 (0.206)	0.013 (0.205)	-0.026 (0.325)	-0.025 (0.323)	-0.027 (0.326)
Main effects						
AI-driven capabilities		0.184** (0.013)	0.201** (0.010)		0.254*** (0.000)	0.136* (0.016)
digital literacy			0.105* (0.021)			
Mediating effect						
GHRM						0.301*** (0.005)
Moderating effect						
AI-driven capabilities X digital literacy			0.270*** (0.002)			
Model summary						
R ²	0.146	0.235	0.302	0.175	0.263	0.372
△R ²		0.089***	0.067***		0.088***	0.109***
F value	5.537***	7.495***	11.943***	4.537***	9.741***	10.480***

Notes: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. 2-tailed. N = 310
^eLog taken

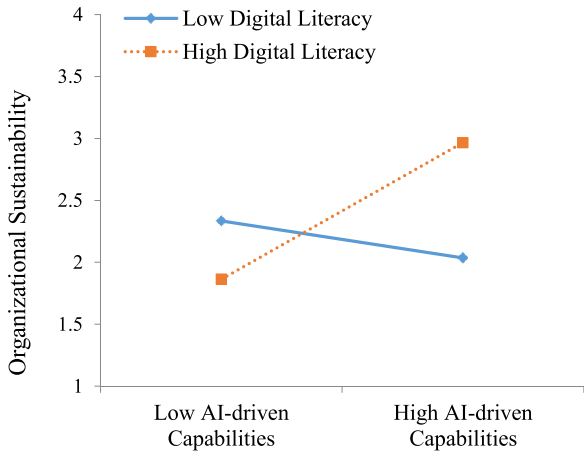


Fig. 2. The moderating effect of digital literacy.

Table III
Conditional indirect effect of AI capabilities on organizational sustainability under digital literacy.

Conditional indirect effect of digital literacy ± 1 SD				
Moderator	Level	Boot indirect effect	Boot SE	95 % bias corrected CI
Digital literacy	Low - 1 SD	0.039	0.030	[-0.033, 0.125]
	Medium	0.053	0.022	[0.038, 0.172]
	High	0.215	0.048	[0.069, 0.364]
	+ 1 SD			

Note. SE: standard error, original size = 310, bootstrapping size = 5000

5. Discussion

Grounded in the DCV, this study explores how AI capabilities contribute to OS under the mediating role of GHRM and the moderating influence of digital literacy. The findings reveal a positive relationship

between AI capabilities and OS. Additionally, GHRM is shown to mediate this link, while digital literacy serves as a moderator. Overall, the study contributes to the digitalization and sustainable development literature by uncovering the underlying mechanisms that drive the relationship between AI capabilities and OS.

5.1. Research findings

The findings of this study offer substantial empirical support for the postulated hypotheses. First, the results show that AI capabilities significantly enhance OS across economic, environmental, and social dimensions. This suggests that firms with stronger AI capabilities are better positioned to integrate sustainable practices into their operations, optimize resource use, and create long-term value for diverse stakeholders. These results align with the DCV, which posits that organizations capable of sensing, seizing, and transforming resources through advanced technologies can better adapt to evolving environmental and societal expectations (Ortiz-Avram et al., 2024). AI systems enable firms to analyze large volumes of data, predict sustainability risks, automate green operations, and innovate sustainable solutions (Al-Raeei, 2025), thereby directly improving their sustainability outcomes.

Second, the findings indicate that GHRM plays a key mediating role in the relationship between AI capabilities and organizational sustainability. This finding indicates that the positive impact of AI capabilities on sustainability is partially channeled through the development and implementation of GHRM practices, such as green recruitment, green training, and green performance management. Firms with robust AI systems are better equipped to design and support these green HRM initiatives, embedding sustainability principles deeper into the organization’s culture and employee behaviors. This is consistent with prior research suggesting that HRM practices are crucial levers for translating technological capabilities into meaningful organizational outcomes (Ali et al., 2025), particularly in the sustainability domain.

Finally, the MODMED analysis further reveals that digital literacy significantly strengthens the indirect link between AI capabilities and OS via GHRM. This suggests that firms where employees and managers possess higher levels of digital literacy are better able to leverage AI technologies in support of green HRM initiatives and, in turn,

sustainability performance. Digital literacy enables employees to effectively use AI tools for sustainable decision-making, data-driven environmental management, and continuous learning regarding sustainable practices (Gasmi et al., 2024). These findings are in line with arguments in the HRD and digital transformation literature that emphasize the role of digital skills in unlocking the full potential of advanced technologies for organizational change and strategic sustainability goals (Ogbeibu et al., 2024; Trujillo-Gallego et al., 2022).

5.2. Theoretical contributions

This study offers three novel contributions to the existing literature in different contexts. First, this study broadens the theoretical and empirical understanding of how AI capabilities contribute to OS. Although prior research acknowledges AI's impact on firm performance (Shaik et al., 2023), innovation (Dong et al., 2024), competitive advantage (Kemp, 2024), and productivity (Czarnitzki et al., 2023), its role in driving sustainability across environmental, economic, and social dimensions remains underexplored. By addressing this gap, our study provides critical insights into how AI capabilities can support organizations' sustainability objectives in a more integrated manner. The findings suggest that AI's ability to process complex environmental data, optimize resource utilization, and enhance operational efficiencies allows firms to meet sustainability goals without sacrificing productivity (Zechiel et al., 2024). Moreover, this contribution deepens our understanding of AI as a transformative tool that shapes sustainability practices beyond traditional efficiency or profitability metrics, helping firms achieve balanced and lasting value across all dimensions of sustainability. Ultimately, these findings highlight AI's crucial role in promoting responsible corporate behavior and positioning business practices to align with long-term sustainability goals.

Second, this study addresses a critical gap between AI capabilities and OS by uncovering the mediating role of GHRM. While existing research underscores AI's potential in the broader sustainability landscape (Kar et al., 2022; Nishant et al., 2020), the mechanisms by which AI capabilities lead to sustainable outcomes are not well understood. Our study advances this understanding by identifying GHRM as a key pathway, suggesting that AI capabilities may not directly drive sustainable outcomes but instead do so by fostering GHRM practices that align employees with environmental objectives. By embedding eco-friendly practices into recruitment, training, and engagement (Ogbeibu et al., 2024), GHRM translates AI-driven efficiencies into actionable sustainability initiatives across the organization. This finding highlights the importance of GHRM as a relational mechanism, amplifying AI's impact on environmental, economic, and social performance. Additionally, by revealing the mediating role of GHRM, we complement previous studies by highlighting its value in linking AI capabilities to OS (Yong et al., 2020). Ultimately, by identifying GHRM as a vital conduit, we contribute a nuanced perspective to the literature on AI and sustainability, underscoring the need for strategic HR practices to fully leverage AI's impact on long-term OS.

Finally, this study advances digital literacy literature within the sustainability context by identifying digital literacy's pivotal role in moderating the indirect effect of AI capabilities on OS via GHRM. Our findings reveal that while AI capabilities enhance sustainability by supporting GHRM practices, this impact is amplified in firms with high digital literacy. This suggests that a digitally literate workforce can better leverage AI-driven efficiencies, translating them into actionable sustainability practices across the organization. By combining mediation and moderation analyses (Preacher et al., 2007), we provide a more comprehensive understanding of how AI capabilities drive sustainable outcomes, emphasizing that both technological capabilities and digital skills are essential to maximizing AI's potential for sustainability. This contribution addresses a gap in the literature, underscoring that digital literacy not only facilitates GHRM but also strengthens the AI-sustainability link, offering a nuanced perspective on the alignment

of digital competencies and AI-driven innovations for impactful, long-term OS. Furthermore, our study highlights that digital literacy equips employees with the skills necessary to adapt to AI-enabled processes (Zahoor et al., 2023), fostering a culture of innovation and adaptability. These insights stress that digital literacy should be prioritized as a vital part of strategic sustainability efforts in AI-enhanced organizations.

5.3. Managerial implications

This study provides critical managerial insights for firms aiming to effectively deploy AI capabilities to enhance OS. First, the findings suggest that managers should not merely view AI capabilities as operational tools but rather as strategic assets for embedding sustainability throughout their organizations. AI should be leveraged to predict environmental risks, optimize energy and resource use, enhance transparency in supply chains, and support ethical labor practices (Abulibdeh et al., 2024). For example, predictive AI models can anticipate future regulatory changes related to carbon emissions, allowing firms to proactively adapt and maintain compliance while enhancing their green credentials (Dong et al., 2024). Firms can also use AI-driven IoT systems to monitor energy consumption across facilities in real-time, making data-driven decisions that simultaneously cut costs and minimize environmental impact (Belhadi et al., 2024). Beyond operational efficiencies, AI can be harnessed to promote social sustainability by using natural language processing to detect employee well-being trends and proactively address workforce challenges (Camilleri, 2024). Managers are therefore encouraged to adopt a mindset where AI is fully integrated into strategic sustainability planning, contributing to long-term resilience, innovation, and competitive advantage.

Second, this study highlights that GHRM practices serve as a crucial catalyst for translating AI capabilities into effective OS outcomes. Managers should actively redesign HR practices to complement AI initiatives, ensuring that sustainability goals are embedded across all stages of the employee lifecycle—from recruitment to development, performance management, and rewards. For instance, organizations could prioritize hiring individuals with expertise in environmental stewardship and digital innovation, and integrate green competencies as core criteria in recruitment and selection processes. Beyond recruitment, managers should design training programs that equip employees with the skills to apply AI tools for sustainable innovation (Al-Swidi et al., 2021), such as using AI for waste reduction, eco-design, or optimizing energy management systems. Moreover, designing green-oriented reward systems is essential to reinforce desired behaviors. Firms could implement incentives such as bonuses, promotions, or public recognition for employees and teams that successfully leverage AI to achieve sustainability targets (e.g., reducing emissions, enhancing diversity and inclusion, promoting ethical sourcing). In doing so, GHRM does not simply react to AI deployment but actively shapes how AI is used to drive environmental, social, and economic performance. Building these synergies ensures that technological investments yield broader systemic changes aligned with corporate sustainability strategies.

Finally, the findings stress that advancing digital literacy is critical for fully realizing the sustainability benefits of AI capabilities through GHRM initiatives. Managers should go beyond basic digital training by fostering a "sustainability-oriented digital culture" across the organization. This involves equipping employees not just with the technical know-how to operate AI tools, but also with the critical thinking skills to apply them ethically and innovatively in the context of environmental and social challenges. For example, digital literacy programs could include modules on using AI for carbon footprint measurement, data ethics in environmental reporting, or developing green innovations with machine learning tools (Abulibdeh et al., 2024). Furthermore, managers should encourage interdisciplinary digital literacy, where HR professionals, sustainability officers, IT experts, and operational teams co-develop AI solutions for sustainability challenges. Investing in such

multifaceted digital skill-building ensures that firms are not just adopting technology for technology's sake, but are fostering an agile, responsible workforce capable of driving sustainable transformation from within.

5.4. Limitations and future avenues

While this study makes novel contributions, it also has limitations that future research should address. First, the cross-sectional design restricts our ability to establish causal relationships. Despite using multiple informants and applying statistical remedies to mitigate this limitation, future studies could benefit from a longitudinal approach to explore causality more effectively. Second, this study examines only digital literacy as a moderator. While valuable, other factors, like the CEO's digital orientation and expertise, may also shape the relationship between AI capabilities and sustainability. Future research should explore these additional moderating factors to uncover further mechanisms that influence sustainable outcomes, broadening our understanding of how leadership and organizational factors interact in an AI-driven context.

Third, this study contributes by highlighting GHRM as a mediator in the link between AI capabilities and OS. However, future research should explore organizational efficiency and agility as alternative mediators, as these elements play a crucial role in adapting to dynamic market conditions and maximizing the sustainable impact of AI capabilities. Examining these factors could provide a more comprehensive understanding of how AI-driven capabilities contribute to sustainability. Finally, the sample, drawn from emerging economies such as Chinese firms, may limit the generalizability of the findings to other economic or institutional contexts. Future research should include diverse institutional settings and economies to improve the robustness and broader applicability of these results.

6. Conclusion

This study advances our understanding of how AI capabilities contribute to OS by uncovering both the pathways and boundary conditions that shape this relationship. Drawing on dynamic capability theory, our findings reveal that AI capabilities significantly enhance sustainability across economic, environmental, and social dimensions,

positioning them as critical enablers of sustainable development in contemporary organizations. Importantly, the results demonstrate that GHRM serves as a key mechanism through which AI capabilities translate into broader sustainability outcomes, highlighting the necessity of embedding environmentally responsible HR practices to fully leverage technological advancements. Furthermore, digital literacy emerges as a vital contextual factor that amplifies the effectiveness of GHRM in mediating the AI-sustainability link, suggesting that organizations must foster a digitally skilled workforce to maximize the sustainability benefits of AI investments.

By offering empirical evidence on the moderated mediation model, this study enriches both the artificial intelligence and human resource development literatures within the sustainability context. It underscores the importance of strategically aligning AI initiatives with human-centered practices and digital skill development to achieve more holistic and enduring sustainability outcomes. These insights not only contribute to academic scholarship but also offer practical guidance for managers seeking to drive sustainable organizational transformation in an increasingly digital and environmentally conscious world.

CRediT authorship contribution statement

Qian Yang: Supervision, Resources. **Adnan Ali:** Writing – review & editing, Validation, Methodology, Conceptualization, Visualization, Software, Investigation. **Muhammad Noman:** Project administration, Investigation, Methodology, Data curation. **Afzaal Ali:** Writing – review & editing, Software, Supervision, Resources. **Zeeshan Ali:** Investigation, Methodology, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors would like to thank the National Natural Science Foundation of China (21&ZD137) for financial support.

Appendix A. Description of the firm (N = 310)

Sample characteristics	Frequency	%	Sample characteristics	Frequency	%
Firm age			Firm size		
0–10	97	31.29	0–100	90	29.03
11–20	127	40.97	100–200	50	16.13
21–30	50	16.13	200–350	35	11.29
31–40	14	04.52	350–500	90	29.03
41–50	12	03.87	Above 500	45	14.52
Above 50	10	03.23	Education		
Gender			High school	15	04.84
Male	233	75.00	Bachelor degree	47	15.16
Female	77	25.00	Master degree	198	63.87
Revenues (in million RMB)			PhD	55	17.74
0–20	75	24.19	Firm regions		
11–40	54	17.42	Eastern	131	42.25
31–60	43	13.87	Western	86	27.75
51–80	15	04.84	Central	93	30.00
51–100	61	19.68	Industry		
Above 100	62	20.00	IT	26	08.39
Ownership			Energy	42	13.55
State-owned	186	60.0	Chemical	68	21.94
Private	97	31.29	Machine	115	37.10
Others	27	08.71	Transportation	25	08.06
			Agriculture	34	10.97

Appendix B. Measurement scales

Constructs	Description	Loadings
AI-driven capabilities	$\alpha = 0.901$; CR = 0.940; AVE = 0.759	
	1. We possess the infrastructure and skilled resources to apply AI information processing system	0.810
	3. We use AI tools to forecast and predict environmental behavior.	0.872
	5. We develop statistical, self-learning, and prediction using AI tools.	0.913
	7. We use AI tools at all levels of business units.	0.830
Organizational sustainable	9. We use AI tools to support the development of new products and services.	0.902
	Economic performance $\alpha = 0.882$; CR = 0.940; AVE = 0.759	
	1. Decrease in cost of materials purchased	0.853
	2. Decrease in cost of energy consumption	0.851
	3. Decrease in fee for waste discharge	0.843
	4. Improvement in return on investment	0.923
	5. Improvement in earnings per share	0.882
	Environmental performance $\alpha = 0.895$; CR = 0.944; AVE = 0.772	
	1. Reduction in air emission	0.924
	3. Reduction in waste (water and/or solid)	0.901
	5. Decrease in consumption of hazardous/harmful/toxic materials	0.852
	7. Decrease in frequency for environmental accidents	0.798
	9. Increase in energy saved due to conservation and efficiency improvements	0.911
	Social performance $\alpha = 0.89$; CR = 0.932; AVE = 0.735	
	1. Improvement in overall stakeholder welfare or betterment	0.822
Green HRM	3. Improvement in community health and safety	0.800
	5. Reduction in environmental impacts and risks to general public	0.924
	7. Improvement in occupational health and safety of employees	0.921
	9. Improved awareness and protection of the claims and rights of people in community served	0.810
	Green analysis and job description $\alpha = 0.879$; CR = 0.887; AVE = 0.724	
	1. Enable involvement in managing environmental activities.	0.820
	2. Enable acquisition of knowledge on environmental management.	0.890
	3. Demanding knowledge on environmental management.	0.842
	Green recruitment $\alpha = 0.868$; CR = 0.826; AVE = 0.704	
	1. The environmental performance of a company attracts new employees	0.813
	2. The company prefers to hire employees who have environmental knowledge	0.864
	Green selection $\alpha = 0.846$; CR = 0.862; AVE = 0.757	
	1. Employee selection takes environmental motivation into account.	0.861
	2. All selection steps consider environmental questions.	0.879
	Green training $\alpha = 0.835$; CR = 0.929; AVE = 0.813	
Digital literacy	1. Environmental training is continuous.	0.931
	2. Environmental training is a priority.	0.924
	3. Environmental training is an important investment.	0.847
	Green performance assessment $\alpha = 0.879$; CR = 0.900; AVE = 0.750	
	1. Every employee has specific environmental goals to achieve.	0.814
	2. Contributions to environmental management are assessed.	0.892
	3. Individual performance assessment results are recorded.	0.889
	Green rewards $\alpha = 0.898$; CR = 0.908; AVE = 0.831	
	1. Cash rewards are provided to recognize environmental performance.	0.892
	2. Environmental performance is recognized publicly	0.931
	Information literacy: $\alpha = 0.843$; CR = 0.890; AVE = 0.730	
	I am confident in	
	1. ... browsing, searching and filtering data, information and digital content	0.846
	2. ... evaluating data, information and digital content	0.832
	3. ... managing data, information and digital content	0.884
	Interaction and collaboration: $\alpha = 0.902$; CR = 0.921; AVE = 0.745	
	I can	
	1. ... interact through digital technologies	0.836
	2. ... share through digital technologies	0.858
	3. ... engage in citizenship through digital technologies	0.931
	4. ... collaborate through digital technologies	0.824
	Digital content creation: $\alpha = 0.883$; CR = 0.881; AVE = 0.712	
	I can	
	1. ... develop digital content	0.801
	2. ... integrate and re-elaborate digital content	0.835
	3. ... respect copyright and licenses	0.892
	Safety: $\alpha = 0.837$; CR = 0.939; AVE = 0.794	
	I am aware of	
	1. ... protecting devices	0.867
	2. ... protecting personal data and privacy	0.866
	3. ... protecting health and well-being	0.910
	4. ... protecting the environment	0.920
	Problem solving: $\alpha = 0.895$; CR = 0.941; AVE = 0.798	
	I am able to	
	1. ... solve technical problems	0.872
	2. ... identify needs and technological responses	0.912
	3. ... creatively use digital technologies	0.914
	4. ... identify digital competence gaps	0.875

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

- Abulibdeh, A., Zaidan, E., & Abulibdeh, R. (2024). Navigating the confluence of artificial intelligence and education for sustainable development in the era of industry 4.0: Challenges, opportunities, and ethical dimensions. *Journal of Cleaner Production*, Article 140527.
- Ahmad, T., Zhang, D., Huang, C., Zhang, H., Dai, N., Song, Y., & Chen, H. (2021). Artificial intelligence in sustainable energy industry: Status Quo, challenges and opportunities. *Journal of Cleaner Production*, 289, Article 125834.
- Aiken, L. S., & West, S. G. (1991). Multiple regression. *Testing and interpreting interactions*. Newbury Park, London Sage.
- Ali, N., Milang, S. W., Sagaf, F. F. A., & Ladjin, N. (2025). Digital transformation in green HRM: Implementation of HR analytics to improve HR sustainability in manufacturing companies in Bandung city. *Journal of Management*, 4(1), 52–64.
- Alnasser, E. M., Alkhozaim, S. M., Alshiha, A. A., Al-Romeedy, B. S., & Khairy, H. A. (2024). Intellectual capital and organisational resilience in tourism and hotel businesses: do organisational agility and innovation matter? *Current Issues in Tourism*, 1–21.
- Al-Raei, M. (2025). The smart future for sustainable development: Artificial intelligence solutions for sustainable urbanization. *Sustainable Development*, 33(1), 508–517.
- Al-Swidi, A. K., Gelaidan, H. M., & Saleh, R. M. (2021). The joint impact of green human resource management, leadership and organizational culture on employees' green behaviour and organisational environmental performance. *Journal of Cleaner Production*, 316, Article 128112.
- Anderson, J. C., & Gerbing, D. W. (1988). Structural equation modeling in practice: A review and recommended two-step approach. *Psychological Bulletin*, 103(3), 411–411.
- Ayoub, H.S., & Aljuhmani, H.Y. (2024). Artificial intelligence capabilities as a catalyst for enhanced organizational performance: The importance of cultivating a data-driven culture. In *Achieving Sustainable Business Through AI, Technology Education and Computer Science: Volume 2: Teaching Technology and Business Sustainability* (pp. 345–356). Springer.
- Baron, R. M., & Kenny, D. A. (1986). The moderator–mediator variable distinction in social psychological research: Conceptual, strategic, and statistical considerations. *Journal of Personality and Social Psychology*, 51(6), 1173–1173.
- Bedué, P., & Fritzsche, A. (2022). Can we trust AI? an empirical investigation of trust requirements and guide to successful AI adoption. *Journal of Enterprise Information Management*, 35(2), 530–549.
- Belhadi, A., Mani, V., Kamble, S. S., Khan, S. A. R., & Verma, S. (2024). Artificial intelligence-driven innovation for enhancing supply chain resilience and performance under the effect of supply chain dynamism: An empirical investigation. *Annals of Operations Research*, 333(2), 627–652.
- Benzidia, S., Makaoui, N., & Bentahar, O. (2021). The impact of big data analytics and artificial intelligence on green supply chain process integration and hospital environmental performance. *Technological Forecasting and Social Change*, 165, Article 120557.
- Berry, J. (1980). Introduction to methodology. *Handbook of Cross-Cultural Psychology*, 2, 1–28.
- Bianchi, G., Testa, F., Boiral, O., & Iraldo, F. (2022). Organizational learning for environmental sustainability: Internalizing lifecycle management. *Organization Environment*, 35(1), 103–129.
- Bocken, N. M., & Geradts, T. H. (2020). Barriers and drivers to sustainable business model innovation: Organization design and dynamic capabilities. *Longest Range Planning*, 53(4), Article 101950.
- Budhwar, P., Malik, A., De Silva, M. T., & Thevisuthan, P. (2022). Artificial intelligence–challenges and opportunities for international HRM: a review and research agenda. *The International Journal of Human Resource Management*, 33(6), 1065–1097.
- Camilleri, M. A. (2024). Artificial intelligence governance: Ethical considerations and implications for social responsibility. *Expert Systems*, 41(7), Article e13406.
- Carayannis, E. G., Sindakis, S., & Walter, C. (2015). Business model innovation as lever of organizational sustainability. *The Journal of Technology Transfer*, 40, 85–104.
- Cetindamar, D., Abedin, B., & Shirahada, K. (2021b). The role of employees in digital transformation: A preliminary study on how employees' digital literacy impacts use of digital technologies. *IEEE Transactions on Engineering Management*.
- Chang, Y.-L., & Ke, J. (2024). Socially responsible artificial intelligence empowered people analytics: A novel framework towards sustainability. *Human Resource Development Review*, 23(1), 88–120.
- Chari, A., Niedenzu, D., Despeisse, M., Machado, C. G., Azevedo, J. D., Boavida-Dias, R., & Johansson, B. (2022). Dynamic capabilities for circular manufacturing supply chains—Exploring the role of Industry 4.0 and resilience. *Business Strategy and the Environment*, 31(5), 2500–2517.
- Chatzopoulou, E.-C., Manolopoulos, D., & Agapitou, V. (2022). Corporate social responsibility and employee outcomes: interrelations of external and internal orientations with job satisfaction and organizational commitment. *Journal of Business Ethics*, 179(3), 795–817.
- Cheung, G. W., & Lau, R. S. (2017). Accuracy of parameter estimates and confidence intervals in moderated mediation models: A comparison of regression and latent moderated structural equations. *Organizational Research Methods*, 20(4), 746–769.
- Chin, Y. S., Mohamad, A. A., & Lo, M. C. (2024). Harnessing the power of artificial intelligence (AI): A paradigm shift in HRM practices for employee sustainable performance. *Global Knowledge, Memory and Communication*.
- Chowdhury, S., Dey, P., Joel-Edgar, S., Bhattacharya, S., Rodriguez-Espindola, O., Abadie, A., & Truong, L. (2023). Unlocking the value of artificial intelligence in human resource management through AI capability framework. *Human Resource Management Review*, 33(1), Article 100899.
- Comrey, A.L., & Lee, H.B. (1992). Interpretation and application of factor analytic results. Comrey AL, Lee HB. A first course in factor analysis, 2, 1992-1992..
- Czarnitzki, D., Fernández, G. P., & Rammer, C. (2023). Artificial intelligence and firm-level productivity. *Journal of Economic Behavior Organization*, 211, 188–205.
- Demir, A., Budur, T., Omer, H. M., & Heshmati, A. (2023). Links between knowledge management and organisational sustainability: does the ISO 9001 certification have an effect? *Knowledge Management Research Practice*, 21(1), 183–196.
- Dey, P. K., Chowdhury, S., Abadie, A., Vann Yaroson, E., & Sarkar, S. (2024). Artificial intelligence-driven supply chain resilience in Vietnamese manufacturing small-and medium-sized enterprises. *International Journal of Production Research*, 62(15), 5417–5456.
- Dohale, V., Akarte, M., Gunasekaran, A., & Verma, P. (2024). Exploring the role of artificial intelligence in building production resilience: Learnings from the COVID-19 pandemic. *International Journal of Production Research*, 62(15), 5472–5488.
- Dong, Z., Xin, Z., Liu, D., & Yu, F. (2024). The impact of artificial intelligence application on company environmental investment in Chinese manufacturing companies. *International Review of Financial Analysis*, 95, Article 103403.
- Fan, D., Xiao, C., Zhang, X., & Guo, Y. (2021). Gaining customer satisfaction through sustainable supplier development: The role of firm reputation and marketing communication. *Transportation Research Part E: Logistics and Transportation Review*, 154, Article 102453.
- Fischer, D., Brettel, M., & Mauer, R. (2020). The three dimensions of sustainability: A delicate balancing act for entrepreneurs made more complex by stakeholder expectations. *Journal of Business Ethics*, 163, 87–106.
- Fornell, C., & Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research*, 18(1), 39–50.
- Gasmi, F., Um, P. N., Virto, L. R., & Saba, P. (2024). Digital literacy, sustainable development and radiation regulation: policy and information systems implications. *Information Systems Frontiers*, 1–31.
- Gupta, S., Drave, V. A., Dwivedi, Y. K., Baabdullah, A. M., & Ismagilova, E. (2020). Achieving superior organizational performance via big data predictive analytics: A dynamic capability view. *Industrial Marketing Management*, 90, 581–592.
- Hair, J.F., Anderson, R.E., Tatham, R.L., & William, C. (1998). *Multivariate data analysis*. Upper Saddle River, NJ: Prentice Hall.
- Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. (2014). *Multivariate data analysis* (7th Edn Pearson new international edition). Harlow, Essex: Pearson Education.
- Hao, X., & Demir, E. (2024). Artificial intelligence in supply chain management: enablers and constraints in pre-development, deployment, and post-development stages. *Production Planning Control*, 1–23.
- Hao, F., Ng, W., Aman, A. M., & Zhang, C. (2024). Technology for sustainability: the impact of Avatar-led Green Training on OCBE and green creativity. *International Journal of Contemporary Hospitality Management*.
- Irfan, E., Ali, Y., & Sabir, M. (2022). Analysing role of businesses' investment in digital literacy: A case of Pakistan. *Technological Forecasting and Social Change*, 176, Article 121484.
- Ishaq, M. I., Sarwar, H., Aftab, J., Franzoni, S., & Raza, A. (2024). Accomplishing sustainable performance through leaders' competencies, green entrepreneurial orientation, and innovation in an emerging economy: Moderating role of institutional support. *Business Strategy and the Environment*, 33(2), 1515–1532.
- Jauhar, S. K., Jani, S. M., Kamble, S. S., Pratap, S., Belhadi, A., & Gupta, S. (2024). How to use no-code artificial intelligence to predict and minimize the inventory distortions for resilient supply chains. *International Journal of Production Research*, 62(15), 5510–5534.
- Jerónimo, H. M., Henriques, P. L., de Lacerda, T. C., da Silva, F. P., & Vieira, P. R. (2020). Going green and sustainable: The influence of green HR practices on the organizational rationale for sustainability. *Journal of Business Research*, 112, 413–421.
- Jiang, X., Jiang, F., Sheng, S., & Wang, G. (2021). A moderated mediation model linking entrepreneurial orientation to strategic alliance performance. *British Journal of Management*, 32(4), 1338–1358.
- Kar, A. K., Choudhary, S. K., & Singh, V. K. (2022). How can artificial intelligence impact sustainability: A systematic literature review. *Journal of Cleaner Production*, 376, Article 134120.
- Kemp, A. (2024). Competitive advantage through artificial intelligence: Toward a theory of situated AI. *Academy of Management Review*, 49(3), 618–635.
- Kılıç, C., & Atilla, G. (2024). Industry 4.0 and sustainable business models: An intercontinental sample. *Business Strategy and the Environment*, 33(4), 3142–3166.
- Kumar, M., Raut, R. D., Mangla, S. K., Ferraris, A., & Choubey, V. K. (2024). The adoption of artificial intelligence powered workforce management for effective revenue growth of micro, small, and medium scale enterprises (MSMEs). *Production Planning Control*, 35(13), 1639–1655.
- Laosirhongthong, T., Adebajo, D., & Tan, K. C. (2013). Green supply chain management practices and performance. *Industrial Management Data Systems*, 113(8), 1088–1109.
- Latinovic, Z., & Chatterjee, S. C. (2022). Achieving the promise of AI and ML in delivering economic and relational customer value in B2B. *Journal of Business Research*, 144, 966–974.

- Lee, M. J., Pak, A., & Roh, T. (2024). The interplay of institutional pressures, digitalization capability, environmental, social, and governance strategy, and triple bottom line performance: A moderated mediation model. *Business Strategy and the Environment*.
- Liao, M., Lan, K., & Yao, Y. (2022). Sustainability implications of artificial intelligence in the chemical industry: A conceptual framework. *Journal of Industrial Ecology*, 26(1), 164–182.
- Lindell, M. K., & Whitney, D. J. (2001). Accounting for common method variance in cross-sectional research designs. *Journal of Applied Psychology*, 86(1), 114–114.
- Liu, L., Song, W., & Liu, Y. (2023). Leveraging digital capabilities toward a circular economy: Reinforcing sustainable supply chain management with Industry 4.0 technologies. *Computers Industrial Engineering*, 178, Article 109113.
- Malik, A., Budhwar, P., Mohan, H., & NR, S. (2023). Employee experience—the missing link for engaging employees: Insights from an MNE's AI-based HR ecosystem. *Human Resource Management*, 62(1), 97–115.
- Mazutis, D., Slawinski, N., & Palazzo, G. (2021). A time and place for sustainability: A spatiotemporal perspective on organizational sustainability frame development. *Business Society*, 60(7), 1849–1890.
- Negri, M., Cagno, E., Colicchia, C., & Sarkis, J. (2021). Integrating sustainability and resilience in the supply chain: A systematic literature review and a research agenda. *Business Strategy and the Environment*, 30(7), 2858–2886.
- Nikou, S., De Reuver, M., & Mahboob Kanafi, M. (2022). Workplace literacy skills—how information and digital literacy affect adoption of digital technology. *Journal of Documentation*, 78(7), 371–391.
- Nishant, R., Kennedy, M., & Corbett, J. (2020). Artificial intelligence for sustainability: Challenges, opportunities, and a research agenda. *International Journal of Information Management*, 53, Article 102104.
- Ogbeibu, S., Emelifeonwu, J., Pereira, V., Oseghale, R., Gaskin, J., Sivarajah, U., & Gunasekaran, A. (2024). Demystifying the roles of organisational smart technology, artificial intelligence, robotics and algorithms capability: A strategy for green human resource management and environmental sustainability. *Business Strategy and the Environment*, 33(2), 369–388.
- de Oliveira, R. T., Ghobakhloo, M., & Figueira, S. (2023). In *Industry 4.0 towards social and environmental sustainability in multinationals: Enabling circular economy, organizational social practices, and corporate purpose*. Elsevier, Article 139712.
- Ortiz-Avram, D., Ovcharova, N., & Engelmann, A. (2024). Dynamic capabilities for sustainability: Toward a typology based on dimensions of sustainability-oriented innovation and stakeholder integration. *Business Strategy and the Environment*, 33(4), 2969–3004.
- Pan, S. L., & Nishant, R. (2023). Artificial intelligence for digital sustainability: An insight into domain-specific research and future directions. *International Journal of Information Management*, 72, Article 102668.
- Podsakoff, N. P. (2003). Common method biases in behavioral research: A critical review of the literature and recommended remedies. *Journal of Applied Psychology*, 88(5), 879, 10–1037.
- Preacher, K. J., Rucker, D. D., & Hayes, A. F. (2007). Addressing moderated mediation hypotheses: Theory, methods, and prescriptions. *Multivariate Behavioral Research*, 42 (1), 185–227.
- Qamar, Y., Agrawal, R. K., Samad, T. A., & Chiappetta Jabbour, C. J. (2021). When technology meets people: The interplay of artificial intelligence and human resource management. *Journal of Enterprise Information Management*, 34(5), 1339–1370.
- Reyes, J., Mula, J., & Díaz-Madroño, M. (2023). Development of a conceptual model for lean supply chain planning in industry 4.0: Multidimensional analysis for operations management. *Production Planning Control*, 34(12), 1209–1224.
- Roppelt, J. S., Schuster, A., Greimel, N. S., Kanbach, D. K., & Sen, K. (2025). Towards effective adoption of artificial intelligence in talent acquisition: A mixed method study. *International Journal of Information Management*, 82, Article 102870.
- Selig, J.P., & Preacher, K.J. (2008). Monte Carlo method for assessing mediation: An interactive tool for creating confidence intervals for indirect effects [Computer software]. Available at: <http://quantpsy.org>. In..
- Shaik, A. S., Alshibani, S. M., Jain, G., Gupta, B., & Mehrotra, A. (2023). Artificial intelligence (AI)-driven strategic business model innovations in small-and medium-sized enterprises. Insights on technological and strategic enablers for carbon neutral businesses. *Business Strategy and the Environment*.
- Singh, R. K., Modgil, S., & Shore, A. (2024). Building artificial intelligence enabled resilient supply chain: A multi-method approach. *Journal of Enterprise Information Management*, 37(2), 414–436.
- Smith, P. A. (2012). The importance of organizational learning for organizational sustainability. *The Learning Organization*, 19(1), 4–10.
- Stoyanova, V., & Stoyanov, S. P. (2024). Scaling up sustainability from an operational capability to a dynamic capability: The Case of Royal Bank of Scotland. *Business Society*, 63(3), 572–625.
- Teece, D., Peteraf, M., & Leih, S. (2016). Dynamic capabilities and organizational agility: Risk, uncertainty, and strategy in the innovation economy. *California Management Review*, 58(4), 13–35.
- Teece, D. J., Pisano, G., & Shuen, A. (1997). Dynamic capabilities and strategic management. *Strategic Management Journal*, 18(7), 509–533.
- Trujillo-Gallego, M., Sarache, W., & de Sousa Jabbour, A. B. L. (2022). Digital technologies and green human resource management: Capabilities for GSCM adoption and enhanced performance. *International Journal of Production Economics*, 249, Article 108531.
- Úbeda-García, M., Claver-Cortés, E., Marco-Lajara, B., & Zaragoza-Sáez, P. (2021). Corporate social responsibility and firm performance in the hotel industry. The mediating role of green human resource management and environmental outcomes. *Journal of Business Research*, 123, 57–69.
- Uddin, M. H., Razzak, M. R., & Rahman, A. A. (2023). Sustainable supply chain management practices, dynamic capabilities and competitive advantage: Evidence from Bangladesh ready-made garments industry. *Business Strategy Development*, 6(2), 176–188.
- Usman, M., Rofcanin, Y., Ali, M., Ogbonnaya, C., & Babalola, M. T. (2023). Toward a more sustainable environment: Understanding why and when green training promotes employees' eco-friendly behaviors outside of work. *Human Resource Management*, 62(3), 355–371.
- Vaio, A., Palladino, R., Hassan, R., & Escobar, O. (2020). Artificial intelligence and business models in the sustainable development goals perspective: A systematic literature review. *Journal of Business Research*, 121, 283–314.
- Vuorikari, R., Punie, Y., Gomez, S.C., & Van Den Brande, G. (2016). DigComp 2.0: The digital competence framework for citizens. Update phase 1: The conceptual reference model.
- Wamba, S. F. (2022). Impact of artificial intelligence assimilation on firm performance: The mediating effects of organizational agility and customer agility. *International Journal of Information Management*, 67, Article 102544.
- Xie, X., & Zhu, Q. (2020). Exploring an innovative pivot: How green training can spur corporate sustainability performance. *Business Strategy and the Environment*, 29(6), 2432–2449.
- Yi, Y., & Demirel, P. (2023). The impact of sustainability-oriented dynamic capabilities on firm growth: Investigating the green supply chain management and green political capabilities. *Business Strategy and the Environment*. <https://doi.org/10.1002/bse.3453>
- Yong, J. Y., Yusliza, M. Y., Ramayah, T., Chiappetta Jabbour, C. J., Sehnem, S., & Mani, V. (2020). Pathways towards sustainability in manufacturing organizations: Empirical evidence on the role of green human resource management. *Business Strategy and the Environment*, 29(1), 212–228.
- Zahoor, N., Zopiatas, A., Adomako, S., & Lamprinakos, G. (2023). The micro-foundations of digitally transforming SMEs: How digital literacy and technology interact with managerial attributes. *Journal of Business Research*, 159, Article 113755.
- Zechiel, F., Blaurock, M., Weber, E., Büttgen, M., & Coussement, K. (2024). How tech companies advance sustainability through artificial intelligence: Developing and evaluating an AI x Sustainability strategy framework. *Industrial Marketing Management*, 119, 75–89.